Ilminur Abulikemu

(615) 359-4602 · ilminurablikim@gmail.com · Portfolio · Murfreesboro, TN

EDUCATION

Middle Tennessee State University Aug. 2025 – May. 2027
M.S. in Data Science, GPA: 3.95/4.0 *Murfreesboro, TN*

- Coursework: Data Mining, Predictive Modeling, Applied Statistical Methods, Model Optimization & Deployment
- Award: MTSU International Merit Scholarship (2025–2026)

Shanghai University of Finance and Economics Sep. 2021 – Jun. 2025
B.A. in Data Journalism, GPA: 3.53/4.0 *Shanghai, China*

- Coursework: Computer Programming, Calculus, Data Mining, Statistical Software, Business Data Analysis
- Award: SUFE Inspirational Scholarship (three consecutive years, 2021–2024)

PUBLICATIONS

Zhang, H., **Abulikemu, Y.**, & Zhou, W. (2026). Measurement and Classification of Piezoelectric Actuator Motion Using Heterodyne Interferometry and Large Language Models. *ASME MSEC 2026*.

RESEARCH EXPERIENCE

Research Assistant, Smart Sensing and Robotics Lab Aug. 2025 – Present
Middle Tennessee State University

- Developing a physics-informed digital twin of a piezoelectric actuator by integrating calibrated hysteresis modeling with a Mamba-based network to learn and correct residual errors.
- Designed an LLM-based framework and Transformer classifier to identify actuator motion patterns from heterodyne-interferometry measurements, supporting nanometer-level motion characterization.
- Built an end-to-end data and evaluation pipeline encompassing preprocessing, database management, cross-validation, and hyperparameter tuning; implemented a Python-based PID controller to translate diagnostic outputs into control adjustments.

Research Intern, Guo Lab Aug. 2026 – Oct. 2026
Vanderbilt University Medical Center

- Curated and annotated 20 transcription factor motif profiles to support a transcription factor-informed TWAS study.
- Processed single-cell ATAC-seq data to identify candidate transcription factor-binding sites.

Research Assistant, Neuromarketing & AI Lab Jan. 2026 – July. 2026
Middle Tennessee State University

- Developed a data-processing workflow to consolidate participant-level reaction-time records across experimental conditions, producing analysis-ready datasets for studies involving 700+ participants.
- Established participant-tracking and data-quality procedures to improve the traceability and reproducibility of behavioral research data.

OTHER EXPERIENCE

Data Science Intern, The Journey Home (Non-Profit) Aug. 2026 – Dec. 2026

- Designing a digital intake platform to replace paper-based client intake at a homeless-services nonprofit.
- Developed a structured data model, a privacy-aware data-sensitivity framework, and a phased implementation plan for the digital intake platform.

Graduate Assistant, Jones College of Business IT Services, MTSU May. 2026 – Dec. 2026

- Build and maintain 10+ Power BI dashboards for cross-departmental reporting; develop Python/JS automation tools

PROJECTS

AscendFi | *Python, FastAPI, Claude API, Vue/Nuxt, Supabase* — VandyHacks Mar. 2026

- Built a full-stack AI finance platform with a multi-agent system for risk prediction and debt optimization

Proto Space | *Unity, Meta Quest 3* — HackMT Jan. 2026

- Developed an augmented-reality CAD viewer for Meta Quest 3, enabling users to inspect imported 3D models through hand-gesture interaction.

TECHNICAL SKILLS

Programming	Python (PyTorch, scikit-learn, pandas), R, SQL, JavaScript, Git
ML & Analytics	Transformers, Mamba, LLMs, NLP, Physics-Informed Modeling, Statistical Analysis, Power BI, Tableau
Languages	English (Fluent), Mandarin (Native), Uyghur (Native), Korean (Intermediate)